PRINCIPLE II: AI-ASSISTED LANGUAGE EDUCATION SHOULD BE RESPONSIBLE

Guideline 4

Foster environmental sustainability

About this guideline

Artificial intelligence has immense potential to transform language education, yet it also introduces a range of social and environmental responsibilities. The **environmental footprint** of AI tools, including carbon emissions, water usage and energy demands, has become a pressing concern (Luccioni et al., 2023). Building the necessary infrastructure, such as servers, computers and data centres, contributes heavily to resource consumption and environmental degradation, and the training of large language models also results in significant greenhouse gas emissions. Additionally, data centres generate substantial heat and require advanced cooling systems, leading to intensive water consumption.

Sustainable integration therefore requires calibrated use, rather than routine default use. At the level of classroom practice, educators can reduce unnecessary computation by designing prompts and workflows to maximise the quality of responses while minimising energy use. Another sustainable practice is to evaluate when and why to use AI. For example, when visuals are needed for teaching materials, existing image banks or open-access libraries may be preferable to prompting AI to generate new images. Similarly, for some information-seeking tasks, conventional web searches may entail substantially lower energy costs (10-30 times less energy) than AI-generated content (De Vries, 2023). At the institutional and provider levels, sustainability also depends on transparency and accountability. Encouraging AI vendors to disclose environmental metrics, such as carbon and water use per operation or per user, would enable educators and education systems to better understand and weigh the often-invisible externalities of AI deployment. This perspective is now endorsed across leading research communities in both environmental science and education.

One way to ensure that this potential is met would be to develop competences in effective prompting. The **IDEA framework** (Park & Choo, 2025) presented below helps guide this process, promoting clear and purposeful prompt design and iterative refinement.

> **Remember**
>
> Consider using digital content that is freely accessible to educators, in order to reduce environmental load.

Did you know?

…that (as of 2023) generating a single image with a powerful AI model like Stable Diffusion, DALL·E or Midjourney uses about as much energy as fully charging a modern smartphone. Less complex AI tasks, like text generation, require far less energy (Luccioni et al., 2023).

Did you know?

Language teachers can use the IDEA framework (Park & Choo, 2025) to write effective prompts. This involves four steps: **I**ncluding essential parts, **D**eveloping with clear prompts, **E**valuating the output and refining prompts, and **A**pplying prompts with accountability.

Including essential PARTS means that teachers should structure their prompts by identifying the following elements:

- **Person:** an explicit statement of the expected AI role (e.g., "a teacher of X foreign language")
- **Aim:** an explicit statement of the task, using an active verb (e.g., "design two learning activities")
- **Recipient:** a specification of the target audience (e.g., "for learners aged X with language level Y")
- **Theme:** context for the task, such as style, tone, constraints, and other relevant content (e.g., "design two learning activities about the structure of a narrative; use an example of a tale; include one good and one bad example; provide sentence models")
- **Structure:** an indication of the desired output format (e.g., "in the form of a Word file", or "tasks should be accompanied by correct answers for self-checking").

Developing with CLEAR prompts involves ensuring that prompts used to generate language learning activities or materials are Concise, Logical, Explicit, Adaptive and Restrictive or domain-specific.

The third step, Evaluate and REFINE, requires **R**ephrasing keywords, and **E**xperimenting with context, using a **F**eedback loop to guide the generation of more accurate outputs. This is usually an **I**terative process, and as language teachers navigate through such iterations, they not only ensure the quality of the current interaction but also the future effectiveness of the system. Finally, **E**valuating and verifying the accuracy of responses is needed to ensure that learning outcomes and efficiency standards are met.

By applying these steps responsibly, perhaps using a clearly defined accountability checklist, language teachers can ensure that AI use in language education leads to improvements in the learning experience.

More to explore

https://trails.upf.edu/

References

Luccioni A. S., Viguier S. and Ligozat A. L. (2023), "Estimating the carbon footprint of BLOOM, a 176B parameter language model", *Journal of Machine Learning Research*, Vol. 24, Issue 253, pp. 1-15.

Park J. and Choo S. (2025), "Generative AI prompt engineering for educators: practical strategies", *Journal of Special Education Technology*, Vol. 40, Issue 3, pp. 411-417, available at: https://doi.org/10.1177/01626434241298954.

Vries A. (de) (2023), "The growing energy footprint of artificial intelligence", *Joule*, Vol. 7, Issue 10, pp. 2191-2194.

Competences and reflection questions

4.1 I can understand the environmental consequences of using AI in language education

Descriptor: *How well am I informed about the environmental consequences of AI use in language education?*

REFLECTION QUESTIONS

- How do the tools I use affect people and ecosystems beyond my classroom?
- What invisible costs (energy, data, labour) accompany apparently simple digital actions in my teaching practice?

4.2 I can reduce the impact of AI use in my classroom

Descriptor: *To what extent can I reduce the environmental impact of my AI use?*

REFLECTION QUESTIONS

- When planning lessons, do I consider whether a low-impact or analogue alternative could achieve the same goal?
- How do I plan, refine, and evaluate my prompts to maximise efficiency in AI use?

4.3 I can promote environmental sustainability when using AI

Descriptor: *How effectively do I teach and model environmentally sustainable uses of AI?*

REFLECTION QUESTIONS

- How do I share examples of environmentally or socially responsible practice with colleagues?
- What institutional policies or collaborations could reduce the environmental and social footprint of AI in education?